RAJALAKSHMI ENGINEERING COLLEGE

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D2

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

void findIntersection(int arr1[], int n1, int arr2[], int n2) {
    int i = 0, j = 0;
    int first = 1; // Helper to manage spaces between outputs

    while (i < n1 && j < n2) {
        if (arr1[i] == arr2[j]) {
            if (!first) {
                printf(" ");
            }
            printf("%d", arr1[i]);
            first = 0;
            i++;
            j++;
        } else if (arr1[i] < arr2[j]) {
            i++;
        } else {
            j++;
        }
    }
    printf("\n");
}

void solve() {
    int n1, n2;
    if (scanf("%d", &n1) != 1) return;

```

```
    int *arr1 = (int *)malloc(n1 * sizeof(int));
    for (int i = 0; i < n1; i++) {
        scanf("%d", &arr1[i]);
    }

    if (scanf("%d", &n2) != 1) {
        free(arr1);
        return;
    }

    int *arr2 = (int *)malloc(n2 * sizeof(int));
    for (int i = 0; i < n2; i++) {
        scanf("%d", &arr2[i]);
    }

    findIntersection(arr1, n1, arr2, n2);

    free(arr1);
    free(arr2);
}

int main() {
    int t;
```

Check

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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